Softdel's BACnet/IP Stack

BACnet/IP Stack PR22 Specifications

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Table of Contents

1 INTRODUCTION 5

1.1 PURPOSE 5

1.2 ABBREVIATIONS / TERMS 5

1.3 REFERENCE INPUTS 6

1.4 DOCUMENT OVERVIEW 6

2 GENERAL STACK SPECIFICATIONS 7

2.1 SYSTEM RESOURCES 7

2.1.1 THREAD SPECIFICATIONS 7

2.1.2 TIMER SPECIFICATIONS 7

2.1.3 INTER-TASK COMMUNICATION 7

2.1.4 INTER-TASK SYNCHRONIZATION 7

2.2 MEMORY REQUIREMENTS 8

2.2.1 ROM OR CODE FOOTPRINT 8

2.2.2 RAM 8

2.3 STACK CONFIGURATIONS 9

2.4 BACNET SERVICES 11

2.5 BACNET OBJECTS 11

2.6 BACNET PROPERTIES 12

2.7 BACNET DATA-TYPES 12

2.8 FEATURES 13

2.8.1 UDP PORT RANGE 13

2.8.2 VIRTUAL DEVICES 13

2.8.3 BBMD AND FD 14

2.8.4 APDU SEGMENTATION 14

2.8.5 CHARACTER ENCODING 14

3 B-ASC (APPLICATION SPECIFIC CONTROLLER) PROFILE 15

3.1 OBJECTS SUPPORTED 15

3.2 SERVICES SUPPORTED 15

3.3 BIBBS SUPPORTED 16

4 B-AAC (ADVANCED APPLICATION CONTROLLER) PROFILE 17

4.1 OBJECTS SUPPORTED 17

4.2 SERVICES SUPPORTED 17

4.3 BIBBS SUPPORTED 18

5 B-BC (BUILDING CONTROLLER) PROFILE 20

5.1 OBJECTS SUPPORTED 20

5.2 SERVICES SUPPORTED 20

5.3 BIBBs Supported 22

6 B-OD (OPERATOR DISPLAY) PROFILE 23

6.1 Objects Supported 23
6.2 Services Supported 25
6.3 BIBBs Supported 25

7 B-OWS (OPERATOR WORKSTATION) PROFILE 27

7.1 Objects Supported 27
7.2 Services Supported 29
7.3 BIBBs Supported 30

8 B-AWS (ADVANCED OPERATOR WORKSTATION) PROFILE 31

8.1 Objects Supported 31
8.2 Services Supported 33
8.3 BIBBs Supported 34

9 MISCELLANEOUS 36

10 REFERENCES 37

Table of Figures

FIGURE 1: PURE-IP AND VIRTUAL DEVICE CONCEPTS....13

Table of Tables

TABLE 1: ABBREVIATIONS....6
TABLE 2: REFERENCE INPUTS....6
TABLE 3: ROM REQUIREMENTS....8
TABLE 4: RAM REQUIREMENTS....9
TABLE 5: STACK SPECIFICATIONS....11
TABLE 6: DATA-TYPE SPECIFICATIONS....13
TABLE 7: B-ASC SUPPORTED OBJECTS....15
TABLE 8: B-ASC SUPPORTED SERVICES....16
TABLE 9: B-ASC SUPPORTED BIBBS....16
TABLE 10: B-AAC SUPPORTED OBJECTS....17
TABLE 11: B-AAC SUPPORTED SERVICES....18
TABLE 12: B-AAC SUPPORTED BIBBS....19
TABLE 13: B-BC SUPPORTED OBJECTS....20
TABLE 14: B-BC SUPPORTED SERVICES....21
TABLE 15: B-BC SUPPORTED BIBBS....22
TABLE 16: B-OD SUPPORTED OBJECTS....25
TABLE 17: B-OD SUPPORTED SERVICES....25
TABLE 18: B-OD SUPPORTED BIBBS....26
TABLE 19: B-OWS SUPPORTED OBJECTS....29
TABLE 20: B-OWS SUPPORTED SERVICES....29
TABLE21: B-OWS SUPPORTED BIBBS....30
TABLE 22: B-AWS SUPPORTED OBJECTS....33
TABLE 23: B-AWS SUPPORTED SERVICES....34
TABLE24: B-AWS SUPPORTED BIBBS....35

1 Introduction

Softdel has been blending technology expertise, domain understanding and process capabilities to deliver engineering services to global names in the industrial world. Softdel provides industrial networking and control centric engineering services covering software development, testing, model-based development and electronics design. Softdel has gained domain experience across Process and Factory Automation, Building Automation, and most recently in Automotive Electronics.

This document is related to **Softdel's BACnet/IP Stack Specifications**.

1.1 Purpose

Softdel's BACnet/IP stack is available and distributed as per standard BACnet profile packages. This document provides information on specifications of stack and various functionalities supported in the stack as per standard BACnet profiles.

1.2 Abbreviations / Terms

Abbreviation	Description
APDU	Application Protocol Data Unit
API	Application Program Interface
B/IP or B-IP	BACnet over Internet Protocol
BMS	Building Management System
BVLL	BACnet Virtual Link Layer
COV	Change of Value
DADR	Destination Address
DEV	Device
DNET	Destination Network Number
IP	Internet Protocol
IPC	Inter-Process Communication
ITC	Inter-Task Communication
N/W or n/w	Network
NA or N/A	Not Applicable or Not Available
NPDU	Network Protocol Data Unit
PDU	Protocol Data Unit
UDP	User Datagram Protocol
RAM	Random Access Memory
ROM	Read Only Memory
VD	Virtual Device
VR	Virtual Router

Table 1: Abbreviations

1.3 Reference Inputs

Reference Name	Description
BACnet ANSI/ASHRAE Standard 135-2020	BACnet Protocol Manual 135-2020
BACnet ANSI/ASHRAE Standard 135-2016	BACnet Protocol Standard 135-2016
BACnet ANSI/ASHRAE Standard 135-2012	BACnet Protocol Standard 135-2012
BACnet ANSI/ASHRAE Standard 135-2010	BACnet Protocol Standard 135-2010
ANSI/ASHRAE Standard 135.1-2013	BACnet Protocol Test Standard 135-2013

Table 2: Reference Inputs

1.4 Document Overview

This document contains different chapters that explain details about various functionalities supported in stack, as per standard BACnet profiles. For each BACnet profile, a separate chapter is provided. Brief description of chapter wise details is mentioned below:

1. General Stack Specifications: Stack specifications common to all profiles, as applicable.
2. B-ASC Profile: Stack specifications for B-ASC profile.
3. B-AAC Profile: Stack specifications for B-AAC profile
4. B-BC Profile: Stack specifications for B-BC profile.
5. B-OD Profile: Stack specifications for B-OD profile.
6. B-OWS Profile: Stack specifications for B-OWS profile.
7. B-AWS Profile: Stack specifications for B-AWS profile.
8. Miscellaneous
9. References

Legends used to in this document are mentioned below:

- ✓ Supported and required in respective profile
- × Not supported and not required in respective profile
- ✓ Supported but not required in respective profile

Hereafter the word '**stack**' or '**Stack**' means '**Softdel's BACnet/IP Stack**'.

2 General Stack Specifications

Softdel's BACnet-IP Stack is a **Multi-Threaded Library** that incorporates the functionalities of BACnet protocol. The multi-threaded library comes in the form of a dynamic link library (.dll) on Windows platform and shared object library (.so) on Linux platform.

Complete implementation of B/IP stack is in '**C Language**' and as per the **ANSI/ASHRAE 135-2020** BACnet standard and supports **Protocol Revision 22**.

The stack has been successfully ported on following OS platforms:

1. Windows (32-bit)
2. Windows (64-bit) (WOW64)
3. Linux (32-bit and 64-bit)

2.1 System Resources

Stack uses various OS resources such as threads, timers, IPC, ITC, message queues, etc. In general, stack can be ported on any OS supporting multi-threading or multi-tasking.

B/IP uses IP-UDP for communication with other BACnet devices in network and hence, the OS should support IP-UDP stack for socket communications.

Stack uses dynamic memory allocation and hence, the OS should support MMU (memory management unit) that allows dynamic memory allocation.

Few of the functionalities in stack may require support for 'File-System'. These functionalities are required in few BACnet profile. However, some of these functionalities are optional and any alternate approach can be developed as per vendor requirements.

Stack being a multi-threaded library uses various OS resources for its functioning. Specifications of these system resources are mentioned in this section.

2.1.1 Thread Specifications

The OS-RTOS should support creation of threads or tasks. Depending upon the BACnet functionalities supported, stack may create 8 to 21 threads.

2.1.2 Timer Specifications

The OS-RTOS should support creation of software timers. Depending upon the BACnet functionalities supported, stack may create 3 to 12 timers.

2.1.3 Inter-Task Communication

The OS-RTOS should support inter-task communication mechanisms. Stack uses 'Message-Queue' for ITC. Stack also uses static queues and linked lists (uses dynamic memory allocation) for sharing data between threads within stack.

2.1.4 Inter-Task Synchronization

The OS-RTOS should support inter-task synchronization mechanisms. Stack uses following OS resources for task/thread synchronization.

1. Semaphores (counting)
2. Mutex
3. Events (Windows)
4. Signals (Linux)

2.2 Memory Requirements

Memory requirements of stack will depend on the functionalities (e.g. services, objects) supported in the stack. Various global variables, queues, linked lists, etc. within stack either use static memory (Data or BSS) or use dynamically allocated memory (Heap) or use both memories.

2.2.1 ROM or Code Footprint

Code footprint of stack as per BACnet profiles and OS is mentioned in below table. The sizes mentioned in the table below are in KBs i.e. Kilobytes.

#	BACnet Profile	Windows-32	Linux-64
1	B-ASC	5045	5318
2	B-AAC	6287	6570
3	B-BC	6327	6613
4	B-OD	4923	4841
5	B-AWS	4693	4519

Table 3: ROM Requirements

Note that the values mentioned in above table are approximate and may vary.

2.2.2 RAM

RAM usage of stack per object instance, as per object types, is mentioned in the table below. The sizes mentioned in the table below are in bytes. Note that the values mentioned in the table below are approximate and may vary.

#	Object Type	Required + Optional Properties	
		Windows-32	Linux-64
1	Device	4028	4168
2	Analog Input	3540	3568
3	Analog Output	7804	7840
4	Analog Value	7532	7568
5	Binary Input	4116	4192
6	Binary Output	8424	8504
7	Binary Value	8140	8224
8	Multi State Input	3748	3800
9	Multi State Output	8000	8040
10	Multi State Value	7752	7808

#	Object Type	Required + Optional Properties	
		Windows-32	Linux-64
11	Notification Class	1108	1152
12	Event Enrollment	4592	4664
13	Calendar	1088	1136
14	Schedule	6004	6168
15	File	3456	3504
16	Trend Log	3612	3656
17	Loop	3476	3512
18	Accumulator	3672	3704
19	Character String Value	8772	8824
20	Structured View	1628	1680
21	Network Port	1832	1904

Table 4: RAM Requirements

Note that the sizes in above table does not include size of dynamically allocated memory (Heap), for 'ARRAY[N]' or 'List' type properties. While calculating the size, 80 bytes of character-string data-type is considered.

In addition to object database within stack, RAM is also used for various queues, linked-lists, global variables, etc. within stack.

2.3 Stack Configurations

Stack internally uses various queues and linked lists for various functionalities. Some of these queues/link-lists are static and some are dynamic in nature. The limits for such queues/link-lists are mentioned in this section.

Note that some maximum limits can be changed depending upon available system resources. The limits mentioned in this section are as per stack's standard support.

In remarks column:

'*Configurable at compile time*' means that the values are hardcoded and can be changed only before stack compilation.

'*Configurable before stack initialization*' means that the values can be changed only before stack initialization. These values cannot exceed hardcoded maximum values defined in stack.

'*Configurable at runtime*' means that the values can be changed after stack initialization i.e. at runtime.

#	Specification	Parameters	Default Value	Max Limits/Values	Remarks
1	Maximum packets received from network at once	Receive Queue and Process Queue size	500	500	Configurable at compile time
2	A-side service requests	Initiate Queue size	1000	1000	Configurable before stack initialization

#	Specification	Parameters	Default Value	Max Limits/Values	Remarks
3	A-side BBMD service request	BBMD Initiate Queue size	100	100	Configurable before stack initialization
4	B-side non-blocking execution	Maximum requests processed simultaneously	5 threads	5 threads	Configurable at compile time
5	B-side callbacks If application exceeds this time, stack will send abort to client device.	B-side timeout	10 seconds	10 seconds	Configurable before stack initialization
6	Maximum time to wait for empty node (when queue is full) before any received packet is dropped	Process Queue timeout	10 seconds	10 seconds	Configurable before stack initialization
7	A-side service request, clear initiate queue timeout	Initiate Queue clear timeout	60 seconds	60 seconds	Configurable before stack initialization
8	A-side BBMD request timeout	BBMD A-side	30 seconds	30 seconds	Configurable before stack initialization
9	B-side service notifications to application	Application Notification Queue size	500	500	Configurable before stack initialization
10	BDT, FDT and Register-FD tables	Maximum entries	10	10	Configurable before stack initialization
11	Segmentation	Maximum data handled i.e. size of local buffers	59040 bytes	59040 bytes	Configurable at compile time
12	Segmentation	Maximum segments that can be received	32	32	Configurable at compile time
13	Segmentation	Maximum segments that can be transmitted	128	128	Configurable at compile time
14	Segmentation	Proposed window size	1	127	Configurable before stack initialization
15	Segmentation	Actual window size	1	127	Configurable before stack initialization
16	UDP port range	B/IP UDP port	Value provided when stack is started	0 to 65535	Use of UDP port number ‘0’ (zero) will depend on operating system
17	Maximum objects in device object	‘Object-List’ property size	1 device object	500	Configurable before stack initialization
18	Dynamic Device Binding: Maximum devices that can be bound	‘Device-Address-Binding’ property size	1 self entry	1000	Configurable before stack initialization
19	Maximum COV subscription requests	Maximum limit	100	100	Configurable before stack initialization

#	Specification	Parameters	Default Value	Max Limits/Values	Remarks
	that can be sent (A-side)				
20	For timing related applications of BACnet (Ex: APDU-retry, Trending, Scheduling, etc.)	Resolution	1 second	1 second	Not applicable

Table 5: Stack Specifications

Note that depending upon functionalities supported in stack, the configurable parameters of stack and their values/limits may differ.

2.4 BACnet Services

Stack supports various BACnet service types. The support for services will vary as per selected BACnet profile. Profile specific service support in stack, is mentioned in further chapters of this document.

In general, all service will function as per required BACnet specifications. Application can register callbacks with stack, to receive A-side response data and B-side request data.

2.5 BACnet Objects

Stack supports various BACnet object types. Support here means that, an object can be created in stack with respective object type and supports all required B-side functionalities. The support for object will vary as per selected BACnet profile. Profile specific object support in stack, is mentioned in further chapters of this document.

Support for proprietary object type can be added as per vendor's requirement.

In general, all objects will function as per required BACnet specifications. There are object functionalities in which the support is not restricted by standard and the support depends on vendor's implementation. Details of such functionalities are mentioned in this section.

Event-Enrollment object supports internal algorithmic reporting only. Event-Enrollment object's, Event-Parameters property supports only following event types:

- Out of Range
- Command Failure
- Change of State

Schedule object supports both internal and external scheduling as per BACnet profile requirements. Schedule object's, Weekly-Schedule and Exception-Schedule properties i.e. 'BACnetTimeValue' data-type in these properties support only following data-types:

- Null
- Boolean
- Real
- Signed or Integer
- Unsigned
- Enumerated

File object supports only Stream-Access. Support for Record-Access can be provided as per vendor's requirements. Maximum file data that File object supports is 2048 bytes. This limit is configurable at compile time.

Trend-Log object supports internal trending only and following Logging-Types:

- Polled
- COV

Loop objects, 'Controlled-Variable-Reference', 'Manipulated-Variable-Reference' and 'Set point-Reference' properties can be configured only for properties with 'Real' data-type.

Accumulator object uses two proprietary properties (defined by Softdel) for its functioning.

2.6 BACnet Properties

Stack supports various BACnet object properties. In general, all required and/or optional properties as per BACnet specifications are supported. Dependencies i.e. B-side linking between properties is supported. Few more optional properties are also supported in stack.

Support for other optional properties and/or proprietary properties can be added as per vendor's requirement.

2.7 BACnet Data-Types

Stack supports various BACnet data-types. The support for encoding and decoding (i.e. A-side read-write support) of data-types will vary as per selected BACnet profile. Profile specific data-type support in stack, is mentioned in further chapters of this document.

Support for proprietary data-types can be added as per vendor's requirement.

In general, all data-type for properties of any object will be supported.

All primitive data-types will have certain boundary values. Similarly all constructed data-types will have certain limits. The limits are applicable to A-side and/or B-side. Table below provides information on these limits.

Note that some maximum limits can be changed depending upon available system resources. The limits mentioned in this section are as per stack's standard support.

#	Data-Type / Property	Max Limit or Max Value	Remarks
1	Character-String	255 bytes	Configurable at compile time
2	Octet-String	50 bytes	Configurable at compile time
3	Bit-String	80 bits (10 bytes)	Configurable at compile time
4	Time-Stamp	Date-Time	Default Time-Stamp type supported in stack
5	'State-Text' property	255 entries	Configurable before stack initialization
6	'Exception-Schedule' property	10 entries	Configurable before stack initialization
7	'BACnetTimeValue' in 'Exception-Schedule' and 'Weekly-Schedule'	10 entries	Configurable before stack initialization

#	Data-Type / Property	Max Limit or Max Value	Remarks
8	'Alarm-Values' property and 'Fault-Values' property	10 entries	Configurable before stack initialization
9	'Date-List' property	10 entries	Configurable before stack initialization
10	'List-of-Object-Property-References' property	10 entries	Configurable before stack initialization
11	'Recipient-List' property	10 entries	Configurable before stack initialization
12	'Configuration-Files' property	10 entries	Configurable before stack initialization
13	'Active-Cov-Subscriptions' property	100 entries	Configurable before stack initialization

Table 6: Data-Type Specifications

2.8 Features

2.8.1 UDP Port Range

BACnet-IP uses IP-UDP communication to communicate with other BACnet device in network. These BACnet devices can use any suitable UDP port for socket communication.

Stack supports UDP port range from 0 to 65535. The use of any specific UDP port number will depend on its availability with the operating system. Also, the use of UDP port number 0 'zero' may vary as per operating system.

2.8.2 Virtual Devices

A BACnet device is one that possesses a device object and communicates using the procedures specified in BACnet standard. In some instances, however, it is desirable to model the activities of a physical building automation and control device through the use of more than one BACnet device. Each such device is referred to as a "Virtual BACnet Device" which is associated with a unique DNET and DARD pair, i.e. a unique BACnet address.

Stack supports creation of one Pure-IP device and multiple Virtual-Devices. The Pure-IP device serves as a "Virtual Router".

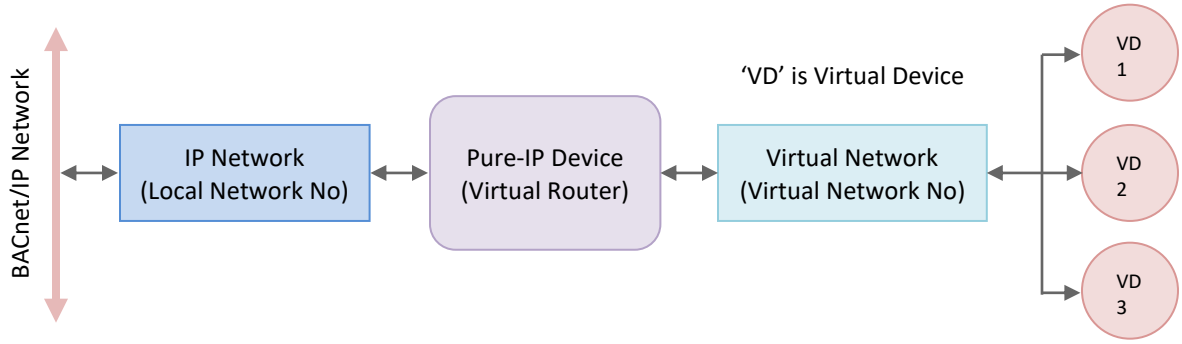

Figure 1: Pure-IP and Virtual Device Concepts

Pure IP Device:

This device supports B/IP communication. Other devices in network can communicate with the device using IP address and UDP port number only.

Virtual Device:

This device supports B/IP communication. Other devices in network can communicate with the device using IP address, UDP port number, network number and MAC address of virtual device.

The pure-IP device acts as a virtual router and takes care of routing the BACnet packets from the BMS system to respective BACnet virtual device.

All the devices created within the stack use single IP address and UDP communication port.

Virtual routing and virtual devices is useful while creating a **BACnet Gateway**. This gateway can be used to map non-BACnet entities into suitable BACnet objects.

Note that virtual routing functionality and virtual device support is not enabled in stack by default. The same can be enabled on vendor's requirement.

2.8.3 BBMD and FD

Stack supports BBMD, Non-BBMD and FD functionalities. Stack can be configured to work either as a BBMD device or as a Non-BBMD device or as a FD. These functionalities can be configured using the APIs provided in stack.

2.8.4 APDU Segmentation

Stack supports APDU segmentation functionality. Support for segmentation is configurable and can be configured at compile-time or at runtime.

- Maximum of 32 segments can be transmitted and the allowed maximum window size is 31.
- Maximum of 32 segments can be received and the allowed maximum window size is 31.

The default value of window size is 1.

2.8.5 Character Encoding

Stack support below mentioned character encoding formats:

1. ANSI X3.4 (ISO 10646 UTF-8)
2. IBM/MS DBCS (code page 850 and 932)
3. ISO 8859-1
4. ISO 10646 (UCS-2)

Note that character string data display or data representation is actually done by application. Stack only encodes and decodes character string data in terms of data bytes.

In case of encoding, stack encodes the character string data bytes as per current character encoding format configured in stack.

In case of decoding, stack decodes the received character string data bytes and character string format. The byte data along with the type of encoding format is sent to application for further processing.

Support for other character sets can be added as per vendor's requirement.

3 B-ASC (Application Specific Controller) Profile

A B-ASC is a controller with limited resources relative to a B-AAC. It is intended for use in a specific application and supports limited programmability.

The B-ASC profile stack supports objects and services mentioned below.

3.1 Objects Supported

BACnet objects supported in B-ASC stack are given in the table below.

#	Object Type	Object Support
1	Device	✓
2	Analog Input	✓
3	Analog Output	✓
4	Analog Value	✓
5	Binary Input	✓
6	Binary Output	✓
7	Binary Value	✓
8	Multi State Input	✓
9	Multi State Output	✓
10	Multi State Value	✓
11	Network Port	✓
12	Bit String Value	✓

Table 7: B-ASC Supported Objects

3.2 Services Supported

BACnet services supported in B-ASC stack are given in below table.

#	Service Classification	Service Description	Initiate	Execute
1	**Alarm and Event**	Subscribe COV	×	✓
2		Confirmed COV Notification	✓	×
3		Unconfirmed COV Notification	✓	×
4	**Object Access**	Read Property	×	✓
5		Write Property	×	✓
6	**Remote Device Management**	Device Communication Control	×	✓
7		Reinitialize Device	×	✓
8		Time Synchronization	×	✓
9		UTC Time Synchronization	×	✓
10		Who-Is	×	✓
11		I-Am	✓	×

#	Service Classification	Service Description	Initiate	Execute
12		Who-Has	×	✓
13		I-Have	✓	×
14	**BBMD**	BBMD, Non-BBMD and FD Functionalities	✓	
15	**Segmentation**	APDU Segmentation Configurable (Tx / Rx / Both / None)	✓	

Table 8: B-ASC Supported Services

3.3 BIBBs Supported

BIBBs supported in B-ASC stack are given in the table below.

#	Supported BIBBs
1	Data Sharing - Read Property – B
2	Data Sharing-Write Property-B
3	Data Sharing – Change Of Value – B
4	Device Management – Dynamic Device Binding – B
5	Device Management – Dynamic Object Binding – B
6	Device Management-Device Communication Control-B

Table 9: B-ASC Supported BIBBs

4 B-AAC (Advanced Application Controller) Profile

A B-AAC is a control device with limited resources relative to a B-BC. It may be intended for specific applications and supports some degree of programmability.

The B-AAC profile stack supports objects and services mentioned below.

4.1 Objects Supported

BACnet objects supported in B-AAC stack are given in below table.

#	Object Type	Object Support
1	Device	✓
2	Analog Input	✓
3	Analog Output	✓
4	Analog Value	✓
5	Binary Input	✓
6	Binary Output	✓
7	Binary Value	✓
8	Multi State Input	✓
9	Multi State Output	✓
10	Multi State Value	✓
11	Accumulator	✓[6]
12	Notification Class	✓
13	Event Enrollment	✓[1]
14	Calendar	✓
15	Schedule	✓[2]
16	Trend Log	✓[4]
17	Structured View	✓
18	Network Port	✓

Table 10: B-AAC Supported Objects

Refer BACnet Objects section for details of objects with ✓[N] symbol.

4.2 Services Supported

BACnet services supported in B-AAC stack are given in the table below.

#	Service Classification	Service Description	Initiate	Execute
1	**Alarm and Event**	Acknowledge Alarm	×	✓
2		Confirmed COV Notification	✓	×
3		Unconfirmed COV Notification	✓	×

#	Service Classification	Service Description	Initiate	Execute
4		Confirmed Event Notification	✓	×
5		Unconfirmed Event Notification	✓	×
6		Get Event Information	×	✓
7		Subscribe COV	×	✓
8	**Object Access**	Create Object	×	✓
9		Delete Object	×	✓
10		Read Property	×	✓
11		Read Property Multiple	×	✓
12		Read Range	×	✓
13		Write Property	×	✓
14		Write Property Multiple	×	✓
15	**Remote Device Management**	Device Communication Control	×	✓
16		Reinitialize Device	×	✓
17		Time Synchronization	×	✓
18		UTC Time Synchronization	×	✓
19		Who-Is	✓	✓
20		I-Am	✓	✓
21		Who-Has	×	✓
22		I-Have	✓	×
23	**BBMD**	BBMD, Non-BBMD and FD Functionalities	✓	
24	**Segmentation**	APDU Segmentation Configurable (Tx / Rx / Both / None)	✓	

Table 11: B-AAC Supported Services

4.3 BIBBs Supported

BIBBs supported in B-AAC stack are given in the table below.

#	Supported BIBBs
1	Data Sharing - Read Property – B
2	Data Sharing - Read Property Multiple– B
3	Data Sharing-Write Property-B
4	Data Sharing-Write Property Multiple-B
5	Data Sharing – Change Of Value – B
6	Scheduling Internal – B
7	Alarm and Event Management – Notification Internal - B
8	Alarm and Event Management – Acknowledge - B

#	Supported BIBBs
9	Alarm and Event Management – Information - B
10	Device Management – Dynamic Device Binding – A
11	Device Management – Dynamic Device Binding – B
12	Device Management – Dynamic Object Binding – B
13	Device Management-Device Communication Control-B
14	Device Management – Time Synchronization – B
15	Device Management – Universal Time Synchronization – B
16	Device Management – Reinitialize Device - B

Table 12: B-AAC Supported BIBBs

5 B-BC (Building Controller) Profile

A B-BC is a general-purpose, field-programmable device capable of carrying out a variety of building automation and control tasks.

The B-BC profile stack supports objects and services mentioned below.

5.1 Objects Supported

BACnet objects supported in B-BC stack are given in the table below.

#	Object Type	Object Support
1	Device	✓
2	Analog Input	✓
3	Analog Output	✓
4	Analog Value	✓
5	Binary Input	✓
6	Binary Output	✓
7	Binary Value	✓
8	Multi State Input	✓
9	Multi State Output	✓
10	Multi State Value	✓
11	Accumulator	✓[6]
12	Notification Class	✓
13	Event Enrollment	✓[1]
14	Calendar	✓
15	Schedule	✓[2]
16	File	✓[3]
17	Trend Log	✓[4]
18	Loop	✓[5]
19	Character String Value	✓
20	Structured View	✓
21	Network Port	✓

Table 13: B-BC Supported Objects

Refer BACnet Objects section for details of objects with ✓[N] symbol.

5.2 Services Supported

BACnet services supported in B-BC stack are given in below table.

#	Service Classification	Service Description	Initiate	Execute
1	**Alarm and Event**	Acknowledge Alarm	×	✓
2		Confirmed COV Notification	✓	×
3		Unconfirmed COV Notification	✓	×
4		Confirmed Event Notification	✓	×
5		Unconfirmed Event Notification	✓	×
6		Get Event Information	×	✓
7		Subscribe COV	×	✓
8	**Audit Reporting**	Audit Log Query	✓	×
9		Confirmed Audit Notification	×	✓
10		Unconfirmed Audit Notification	×	✓
11	**File Access**	Atomic Read File	×	✓
12		Atomic Write File	×	✓
13	**Object Access**	Add List Element	×	✓
14		Remove List Element	×	✓
15		Create Object	×	✓
16		Delete Object	×	✓
17		Read Property	✓	✓
18		Read Property Multiple	✓	✓
19		Read Range	×	✓
20		Write Property	✓	✓
21		Write Property Multiple	✓	✓
22	**Remote Device Management**	Device Communication Control	×	✓
23		Reinitialize Device	×	✓
24		Time Synchronization	×	✓
25		UTC Time Synchronization	×	✓
26		Who-Is	✓	✓
27		I-Am	✓	✓
28		Who-Has	✓	✓
29		I-Have	✓	✓
29	**BBMD**	BBMD, Non-BBMD and FD Functionalities	✓	
30	**Segmentation**	APDU Segmentation Configurable (Tx / Rx / Both / None)	✓	

Table 14: B-BC Supported Services

5.3 BIBBs Supported

BIBBs supported in B-BC stack are given in the table below.

#	Supported BIBBs
1	Data Sharing - Read Property – A
2	Data Sharing - Read Property – B
3	Data Sharing - Read Property Multiple– A
4	Data Sharing - Read Property Multiple– B
5	Data Sharing-Write Property – A
6	Data Sharing-Write Property - B
7	Data Sharing-Write Property Multiple-B
8	Data Sharing – Change Of Value – B
9	Scheduling External – B
10	Alarm and Event Management – Notification Internal - B
11	Alarm and Event Management – Acknowledge - B
12	Alarm and Event Management – Information - B
13	Trending-Viewing and Modifying Trends-Internal-B
14	Trending-Automated Trend Retrieval-B
15	Device Management – Dynamic Device Binding – A
16	Device Management – Dynamic Device Binding – B
17	Device Management – Dynamic Object Binding – B
18	Device Management-Device Communication Control-B
19	Device Management – Time Synchronization – B
20	Device Management – Universal Time Synchronization – B
21	Device Management – Reinitialize Device - B
22	Device Management – Backup and Restore - B

Table 15: B-BC Supported BIBBs

6 B-OD (Operator Display) Profile

The B-OD is a basic operator interface with limited capabilities relative to a B-OWS and B-AWS. It is not intended to perform direct digital control. The B-OD profile could be used for wall-mounted LCD devices, displays affixed to BACnet devices; handheld terminals or other very simple user interfaces.

The B-OD profile stack supports objects and services mentioned below.

6.1 Objects Supported

BACnet objects supported in B-OD stack are given in the table below. Note that only client-side functionalities are supported for these objects.

#	Object Type	Object Support
1	Device	✓
2	Analog Input	✓
3	Analog Output	✓
4	Analog Value	✓
5	Binary Input	✓
6	Binary Output	✓
7	Binary Value	✓
8	Multi State Input	✓
9	Multi State Output	✓
10	Multi State Value	✓
11	Notification Class	✓
12	Event Enrollment	✓
13	Loop	✓
14	Accumulator	✓
15	Command	✓
16	Program	✓
17	Averaging	✓
18	Pulse Converter	✓
19	Load Control	✓
20	Bit String Value	✓
21	Character String Value	✓
22	Date Pattern Value	✓
23	Date Value	✓
24	Date Time Pattern Value	✓
25	Date Time Value	✓
26	Integer Value	✓

#	Object Type	Object Support
27	Large Analog Value	✓
28	Octet String Value	✓
29	Positive Integer Value	✓
30	Time Pattern Value	✓
31	Time Value	✓
32	Channel	✓
33	Lighting Output	✓
34	Binary Lighting Output	✓
35	Network Port	✓
36	Timer	✓
38	Date Pattern Value	✓
39	Date Value	✓
40	Date Time Pattern Value	✓
41	Date Time Value	✓
42	Integer Value	✓
43	Large Analog Value	✓
44	Octet String Value	✓
45	Positive Integer Value	✓
46	Time Pattern Value	✓
47	Time Value	✓
48	Group	✓
49	Network Security	✓
50	Global Group	✓
51	Alert Enrollment	✓
52	Notification Forwarder	✓
53	Channel	✓
54	Lighting Output	✓
55	Binary Lighting Output	✓
56	Network Port	✓
57	Timer	✓
58	Elevator Group	✓
59	Lift	✓
60	Escalator	✓
61	Staging	✓

#	Object Type	Object Support
62	Audit Log	✓
63	Audit Reporter	✓

Table 16: B-OD Supported Objects

6.2 Services Supported

BACnet services supported in B-OD stack are given in the table below.

#	Service Classification	Service Description	Initiate	Execute
1	**Alarm and Event**	Confirmed COV Notification	×	✓
2		Unconfirmed COV Notification	×	✓
3		Confirmed Event Notification	×	✓
4		Unconfirmed Event Notification	×	✓
5		Subscribe COV	✓	×
6	**Object Access**	Read Property	✓	✓
7		Read Property Multiple	✓	×
8		Write Property	✓	✓
9		Write Property Multiple	✓	×
10	**Remote Device Management**	Reinitialize Device	×	✓
11		Who-Is	✓	✓
12		I-Am	✓	✓
13		Who-Has	✓	✓
14		I-Have	✓	✓
15	**BBMD**	BBMD, Non-BBMD and FD Functionalities	✓	
16	**Segmentation**	APDU Segmentation Configurable (Tx / Rx / Both / None)	✓	

Table 17: B-OD Supported Services

6.3 BIBBs Supported

BIBBs supported in B-OD stack are given in the table below.

#	Supported BIBBs
1	Data Sharing - Read Property – A
2	Data Sharing - Read Property – B
3	Data Sharing-Write Property – A
4	Data Sharing-Write Property – B
5	Data Sharing-View – A
6	Data Sharing-Modify – A

#	Supported BIBBs
7	Alarm and Event Management – Notification - A
8	Alarm and Event Management – View Notification - A
9	Device Management – Dynamic Device Binding – A
10	Device Management – Dynamic Device Binding – B
11	Device Management – Dynamic Object Binding – B
12	Device Management – Reinitialize Device - B

Table 18: B-OD Supported BIBBs

7 B-OWS (Operator Workstation) Profile

The B-OWS is an operator interface with limited capabilities relative to a B-AWS. The B-OWS is used for monitoring and basic control of a system but differs from a B-AWS in that it does not support configuration activities, nor does it provide advanced troubleshooting capabilities.

The B-OWS profile is targeted at the daily operator who needs the ability to monitor basic system status and to perform simple modifications to the operation of the system.

The B-OWS profile stack supports objects and services mentioned below.

7.1 Objects Supported

BACnet objects supported in B-OWS stack are given in the table below. Note that only client-side functionalities are supported for these objects.

#	Object Type	Object Support
1	Device	✓
2	Analog Input	✓
3	Analog Output	✓
4	Analog Value	✓
5	Binary Input	✓
6	Binary Output	✓
7	Binary Value	✓
8	Multi State Input	✓
9	Multi State Output	✓
10	Multi State Value	✓
11	Notification Class	✓
12	Event Enrollment	✓
13	Calendar	✓
14	Schedule	✓
15	File	✓
16	Trend Log	✓
17	Loop	✓
18	Accumulator	✓
19	Command	✓
20	Program	✓
21	Life Safety Point	✓
22	Life Safety Zone	✓
23	Averaging	✓

#	Object Type	Object Support
24	Pulse Converter	✓
25	Event Log	✓
26	Trend Log Multiple	✓
27	Load Control	✓
28	Structured View	✓
29	Access Door	✓
30	Access Credential	✓
31	Access Point	✓
32	Access Rights	✓
33	Access User	✓
34	Access Zone	✓
35	Credential Data Input	✓
36	Bit String Value	✓
37	Character String Value	✓
38	Date Pattern Value	✓
39	Date Value	✓
40	Date Time Pattern Value	✓
41	Date Time Value	✓
42	Integer Value	✓
43	Large Analog Value	✓
44	Octet String Value	✓
45	Positive Integer Value	✓
46	Time Pattern Value	✓
47	Time Value	✓
48	Group	✓
49	Network Security	✓
50	Global Group	✓
51	Alert Enrollment	✓
52	Notification Forwarder	✓
53	Channel	✓
54	Lighting Output	✓
55	Binary Lighting Output	✓
56	Network Port	✓

#	Object Type	Object Support
57	Timer	✓
58	Elevator Group	✓
59	Lift	✓
60	Escalator	✓
61	Staging	✓
62	Audit Log	✓
63	Audit Reporter	✓

Table 19: B-OWS Supported Objects

7.2 Services Supported

BACnet services supported in B-OWS stack are given in below table.

#	Service Classification	Service Description	Initiate	Execute
1	**Alarm and Event**	Acknowledge Alarm	✓	×
2		Confirmed COV Notification	×	✓
3		Unconfirmed COV Notification	×	✓
4		Confirmed Event Notification	×	✓
5		Unconfirmed Event Notification	×	✓
6		Get Event Information	✓	×
7		Subscribe COV	✓	×
8	**Object Access**	Read Property	✓	✓
9		Read Property Multiple	✓	×
10		Read Range	✓	×
11		Write Property	✓	✓
12		Write Property Multiple	✓	×
13	**Remote Device Management**	Reinitialize Device	×	✓
14		Time Synchronization	✓	×
15		UTC Time Synchronization	✓	×
16		Who-Is	✓	✓
17		I-Am	✓	✓
18		Who-Has	✓	✓
19		I-Have	✓	✓
20	**BBMD**	BBMD, Non-BBMD and FD Functionalities	✓	
21	**Segmentation**	APDU Segmentation Configurable (Tx / Rx / Both / None)	✓	

Table 20: B-OWS Supported Services

7.3 BIBBs Supported

BIBBs supported in B-OWS stack are given in the table below.

#	Supported BIBBs
1	Data Sharing - Read Property – A
2	Data Sharing - Read Property – B
3	Data Sharing - Read Property Multiple– A
4	Data Sharing-Write Property – A
5	Data Sharing-Write Property – B
6	Data Sharing-View – A
7	Data Sharing-Modify – A
8	Data Sharing-Write Property Multiple-A
9	Scheduling-View and Modify-A
10	Alarm and Event Management – Notification - A
11	Alarm and Event Management – Acknowledge - A
12	Alarm and Event Management-Alarm Summary View-A
13	Alarm and Event Management-View and Modify-A
14	Alarm and Event Management-View Notifications-A
15	Trending-View - A
16	Device Management – Dynamic Device Binding – A
17	Device Management – Dynamic Device Binding – B
18	Device Management – Dynamic Object Binding – B
19	Device Management-Manual Time Synchronization-A
20	Device Management – Reinitialize Device - B

Table21: B-OWS Supported BIBBs

8 B-AWS (Advanced Operator Workstation) Profile

The B-AWS is the advanced operator's window into a BACnet system. It is primarily used to monitor the performance of a system and to modify parameters that affect the operation of a system. It may also be used for configuration activities that are beyond the scope of this standard.

The B-AWS profile is targeted at a building operator or technician with a higher level of technical ability. It provides support for limited configuration actions and ongoing commissioning activities.

The B-AWS profile stack supports objects and services mentioned below.

8.1 Objects Supported

BACnet objects supported in B-AWS stack are given in below table. Note that only client-side functionalities are supported for these objects.

#	Object Type	Object Support
1	Device	✓
2	Analog Input	✓
3	Analog Output	✓
4	Analog Value	✓
5	Binary Input	✓
6	Binary Output	✓
7	Binary Value	✓
8	Multi State Input	✓
9	Multi State Output	✓
10	Multi State Value	✓
11	Notification Class	✓
12	Event Enrollment	✓
13	Calendar	✓
14	Schedule	✓
15	File	✓
16	Trend Log	✓
17	Loop	✓
18	Accumulator	✓
19	Command	✓
20	Program	✓
21	Life Safety Point	✓
22	Life Safety Zone	✓
23	Averaging	✓
24	Pulse Converter	✓

#	Object Type	Object Support
25	Event Log	✓
26	Trend Log Multiple	✓
27	Load Control	✓
28	Structured View	✓
29	Access Door	✓
30	Access Credential	✓
31	Access Point	✓
32	Access Rights	✓
33	Access User	✓
34	Access Zone	✓
35	Credential Data Input	✓
36	Bit String Value	✓
37	Character String Value	✓
38	Date Pattern Value	✓
39	Date Value	✓
40	Date Time Pattern Value	✓
41	Date Time Value	✓
42	Integer Value	✓
43	Large Analog Value	✓
44	Octet String Value	✓
45	Positive Integer Value	✓
46	Time Pattern Value	✓
47	Time Value	✓
48	Group	✓
49	Network Security	✓
50	Global Group	✓
51	Alert Enrollment	✓
52	Notification Forwarder	✓
53	Channel	✓
54	Lighting Output	✓
55	Binary Lighting Output	✓
56	Network Port	✓
57	Timer	✓

#	Object Type	Object Support
58	Elevator Group	✓
59	Lift	✓
60	Escalator	✓
61	Staging	✓
62	Audit Log	✓
63	Audit Reporter	✓

Table 22: B-AWS Supported Objects

8.2 Services Supported

BACnet services supported in B-AWS stack are given in below table.

#	Service Classification	Service Description	Initiate	Execute
1	**Alarm and Event**	Acknowledge Alarm	✓	×
2		Confirmed COV Notification	×	✓
3		Unconfirmed COV Notification	×	✓
4		Confirmed Event Notification	×	✓
5		Unconfirmed Event Notification	×	✓
6		Get Event Information	✓	×
7		Subscribe COV	✓	×
8	**Audit Reporting**	Audit Log Query	✓	×
9		Confirmed Audit Notification	×	✓
10		Unconfirmed Audit Notification	×	✓
11	**File Access**	Atomic Read File	✓	×
12		Atomic Write File	✓	×
13	**Object Access**	Add List Element	✓	×
14		Remove List Element	✓	×
15		Create Object	✓	×
16		Delete Object	✓	×
17		Read Property	✓	✓
18		Read Property Multiple	✓	×
19		Read Range	✓	×
20		Write Property	✓	✓
21		Write Property Multiple	✓	×
22		Write Group	✓	×
23		Device Communication Control	✓	×

#	Service Classification	Service Description	Initiate	Execute
24	**Remote Device Management**	Reinitialize Device	✓	✓
25		Time Synchronization	✓	×
26		UTC Time Synchronization	✓	×
27		Who-Is	✓	✓
28		I-Am	✓	✓
29		Who-Has	✓	✓
30		I-Have	✓	✓
31		Who-Am-I	×	✓
32		You-Are	✓	×
33	**BBMD**	BBMD, Non-BBMD and FD Functionalities	✓	
34	**Segmentation**	APDU Segmentation Configurable (Tx / Rx / Both / None)	✓	

Table 23: B-AWS Supported Services

8.3 BIBBs Supported

BIBBs supported in B-AWS stack are given in the table below.

#	Supported BIBBs
1	Data Sharing - Read Property – A
2	Data Sharing - Read Property – B
3	Data Sharing - Read Property Multiple– A
4	Data Sharing-Write Property – A
5	Data Sharing-Write Property – B
6	Data Sharing-Write Property Multiple-A
7	Data Sharing-Advanced View – A
8	Data Sharing-Advanced Modify – A
9	Scheduling-Advanced View and Modify-A
10	Alarm and Event Management – Notification - A
11	Alarm and Event Management – Acknowledge - A
12	Alarm and Event Management-Alarm Summary View-A
13	Alarm and Event Management-Advanced View and Modify -A
14	Alarm and Event Management-Advanced View Notifications-A
15	Alarm and Event Management-Event Log View and Modify-A
16	Trending- Advanced View and Modify - A
17	Device Management – Dynamic Device Binding – A
18	Device Management – Dynamic Device Binding – B

#	Supported BIBBs
19	Device Management – Dynamic Object Binding – B
20	Device Management-Manual Time Synchronization-A
21	Device Management-Automatic Network Mapping-A
22	Device Management-Automatic Device Mapping-A
23	Device Management-Device Communication Control-A
24	Device Management- Reinitialize Device - A
25	Device Management-Backup and Restore - A
26	Device Management-Object Create and Delete – A
27	Device Management-Dynamic Device Assignment-A
28	Network Management-Communications Configuration-A
29	Audit Reporting-Advanced View and Modify-A
30	Device Management – Reinitialize Device - B

Table24: B-AWS Supported BIBBs

9 Miscellaneous

Not applicable.

10 References

Not applicable.